

Random Variable

Random Variable is a mathematical function that assigns a real number to each possible outcome in a sample space of a random process.

It allow you to model the whole experiment at once.

There are two type of Random Variable:

1. **Discrete Random Variable:** Finite number of values or could be infinite too (countable) ex: the number of heads in 3 coin flips
2. **Continuous Random Variable:** Infinite number of values (uncountable number of values within an interval) ex: height of the a person or the time until a bulb burns out

Random Variable vs Deterministic Variable

Deterministic

$$x = 2, f(x) = x^2$$

Fixed outcome

Random

X = number of defective items in
a shipment

Uncertain outcome